MIL-C-915E
AMENDMENT-1
5 April 1973

MILITARY SPECIFICATION

CABLE AND CORD ELECTRICAL, FOR SHIPBOARD USE

GENERAL SPECIFICATIONS FOR

This amendment forms a part of Military Specification MIL-C-915E, dated 1 August 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 3

2.1: Add the following Military Specification:

"MIL-I-45208 - Inspection System Requirements."

Page 6

3.3.9, first sentence: Delete and substitute: "Tie cords shall be of cotton or synthetic fiber having a maximum diameter of 0.065 inch and a minimum breaking strength of 30 pounds."

3.4.1: Delete and substitute:

"3.4.1 Conductor stranding. The size and quantity of individual conductor strands and the total circular-mil area of the conductor, unless otherwise specified on the specification sheet, shall be in accordance with tables I, II or III, as applicable, in accordance with the conductor size designation indicated on the specification sheet. When AWG sizes are specified on the specification sheets for conductors, they shall be in accordance with ASTM B8."

Page 8

Table III, column 2: Delete all values under "Rope" and substitute the following:

Number of strands Minimum
" 49
228
49
304
336
209
49
532
304
133
2107
418
2646
259
3325
259
648
760
'427
988
259
1254
6384
259
2052
259
427
427
4033"

6145

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Page 10

Table IV: Add the following as color, conductor or group No. "50".

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
"50	Black	Orange	Red"

Page 16

Add the following paragraph as 4.1.1:

"4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208."

Page 20

4.6.1: Delete and substitute:

"4.6.1 Sampling procedure for comparison inspection. Once a manufacturer's product has been qualified, the manufacturer shall submit one specimen for comparison inspection from the first lot of each type and size that he produces. Thenceforth, the manufacturer shall submit one specimen of each type and size that he produces within each calendar year. The entire lot of cable or cord which is represented by the specimen submitted for comparison inspection shall be retained by the manufacturer until he receives, from the Government, notice of the results of comparison inspection."

4.6.1.2: Add the following items (a) through (g):

"(a) Manufacturer's name
"(b) Cable type-and-size designation
"(c) Applicable specification (that is, the MIL-C-915 specification sheet number)
"(d) The test or qualification reference
"(e) Footage represented by the specimen
"(f) The number of the coil or reel from which the specimen was taken
"(g) Either the manufacturing number or the order number assigned by the customer"

Pages 25 and 26

4.8.8 through 4.8.8.3.1: Delete and substitute:

"4.8.8 Cable aging and compatibility (125°C.). This test is applicable to cables having either thermosetting insulation or silicone rubber-glass tape insulation, and is for the purpose of detecting any significant degradation in the characteristics of the cable or its components, resulting from either component incompatibility or prolonged overheating. The cable aging may be accomplished either by the current overload method or by the hot air oven method.

"4.8.8.1 Specimens. The specimen for the current overload method shall consist of a length of completed cable approximately 40 feet long. The specimens for the hot air oven method shall consist of two lengths of completed cable: One length approximately 30 feet long and one length 2 feet plus 24 times the overall diameter of the specimen.

"4.8.8.2 Apparatus. The test apparatus for both methods shall consist of a heat chamber or oven, a means of measuring and controlling the chamber or oven temperature within plus or minus 3°C of the specified temperature for a continuous period of 400 hours, an internal mercury thermometer and a rack or other means of supporting the specimen or specimens within the chamber or oven. For the current overload method there shall be a source of current capable of supplying the required amount of current for a continuous period of 400 hours. For both methods there shall also be a mandrel having a diameter of approximately 12 times the specimen diameter.

"4.8.8.3 Procedure. The procedure for each method shall consist of two parts: (1) heat aging; (2) bending.

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Pages 25 and 26 (Cont'd)

"4.8.8.3.1 Part one - heat aging. (may be accomplished by either of the following methods).

(a) Current overload method: Prior to heat aging, the specimen shall be given the voltage withstand and the insulation resistance tests. The maximum values of insulation resistance measured for each conductor shall be recorded. During aging, conductor temperature shall be determined either by direct measurement or by the measurement of change of copper resistance. For the direct measurement method, a thermocouple shall be inserted through a small knife puncture in the cable, approximately midway between the ends of the specimen. For cables having only one layer of conductors, the thermocouple shall be placed in contact with the strands of any conductor. For cables having more than one layer of conductors, the thermocouple shall be placed in contact with the strands of any conductor in the innermost layer. Thermocouples shall also be attached to the jacket of unarmored cables. In the case of armored cables, the thermocouples shall be attached to the armor. The specimen shall be placed into the chamber so as to have a straight section of approximately 7 feet, and so as to permit free circulation of convection air currents. The chamber temperature shall be maintained at 50° +3°C during the 400-hour period. The amplitudes of loading currents shall be such as to maintain each conductor at a temperature of 125° +5°C. continuously for 400 hours. Readings of time, chamber temperature, jacket or armor temperature, and the current in each conductor shall be recorded at 15-minute intervals during the first hour, at 1-hour intervals during the next 5 hours, and twice daily thereafter for the duration of the test. Record shall also be made of any gas or fumes emanating from the specimen, and of any oozing of material from the ends of the specimen. In the case of armored cable, record shall also be made of any exudation of the jacket material through the armor. At the end of the 400-hour test period, the specimen shall be allowed to cool to room temperature. Then it shall be given the voltage withstand and the insulation resistance tests. The maximum insulation resistance values measured for each conductor shall again be recorded. From the portion of the 40-foot specimen which was kept straight during the heat aging, a specimen having an approximate length of 2 feet plus 24 times the overall diameter of the specimen shall be taken. This specimen shall be used for the bending procedure. From the remainder of the 40-foot specimen, an additional specimen shall be taken and given the cable filler removability test.

(b) Hot air oven method. Prior to heat aging, the 30-foot specimen shall be given the voltage withstand and the insulation resistance tests. The maximum value of insulation resistance measured for each conductor shall be recorded. The two specimens shall be suspended horizontally in the oven, with the shorter specimen kept straight. The oven temperature shall be maintained at 125° +3°C. during the 400-hour period. Record shall be made of any gases or fumes emanating from the specimens and of any oozing of material from the ends of either specimen. In the case of armored cable, record shall also be made of any exudation of jacket material through the armor. At the end of the 400-hour test period, the specimens shall be allowed to cool to room temperature. The shorter specimen shall be used for the bending procedure. The 30-foot specimen shall again be given the voltage withstand and the insulation resistance tests. The maximum value of insulation resistance measured for each conductor shall again be recorded. One additional specimen shall then be taken from the 30-foot specimen, and given the cable filler removability test."

4.8.8.4, items (b), (c), and (d), line 1: Delete "40-foot".

Page 29

4.8.16.2.4, line 3: Delete "certerlines" and substitute "centerlines".

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Pages 34 and 35

4.8.22.3.2, fourth sentence: Delete and substitute: "Each specimen shall be stretched at a uniform rate of 20 inches per minute (unless otherwise specified on the specification sheet), until rupture."

Page 41

4.9.5.1, line 5: After "all other conductors", add "and".

Page 44

4.9.8.2, first sentence: Delete and substitute: "The test voltages shall be as specified in the applicable specification sheet. The point of application shall be conductor to conductor unless otherwise specified on the applicable specification sheet."

Custodians:
Army - MI
Navy - SH
Air Force - 80
Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0648)

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